INCH-POUND
MIL-DTL-24643/22D
22 August 2002
SUPERSEDING
MIL-C-24643/22C
14 March 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 5000 VOLTS, TYPE LS5KVTSGU (INCLUDING VARIATION LS5KVTSGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, (see Table I for size).

Second - Silicone rubber - glass tape insulation (see Table I for diameter). Standard identification code applied by method 1 or letter identification code by method 5.

Third - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly. Cabling sequence shall be A, B and C.

Fourth - A binder or combination binder and barrier.

Construction variant for type LS5KVTSGU

Fifth - Cross-linked polyolefin jacket. Minimum average thickness 0.090 inch.

Construction variant for type LS5KVTSGA

Fifth - Cross-linked polyolefin jacket. Minimum average thickness 0.090 inch.

Sixth - Braided metal armor.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Details.

Military part no. 1/ M24643/22	Size 2/	Conductors size		Type's overall diameter (inches)				Diameter over outer insulation tape nominal (in.)	Conductor resistance per 1000 ft at 25°C max (ohms)	Insulation resistance (megohms)	Cold bending mandrel dia max (in.)	Accele-rated service loading (ampere)
				LS5KVTSGU		LS5KVTSGA						
		Navy std	AWG	min	max	min	max					
-01UN	100		0 (class D)		1.74		1.79	0.675	0.106	400	22	295
-02UN	150		000 (class D)		1.95		2.00	.769	.0674	350	24	400
-03UN	250		250,000 (class C)		2.22		2.27	.889	.0453	300	28	535
-04UN	350		350,000 (class D)		2.45		2.50	.994	.0321	265	30	665
-05UN	400	400 (127)	----		2.60		2.65	1.054	.0273	250	32	750

1/ Unarmored. When armored option is required, see MIL-DTL-24643 for configuration of part number.
2/ Type LS5KVTSGU and LS5KVTSGA.

EXAMINATION AND TESTS:

Basic electrical: Requirements:
- Conductor resistance - ohms/1000 feet at 25, maximum ..... (see Table I)
- Voltage withstand - volts, root mean square, minimum 13,500
- Insulation resistance - megohms -1000 feet, minimum
  - Conductor to conductor ..... (see Table I)
  - Conductor continuity ..... No failure

Group A:
- Visual and dimensional ..... No failure
- Watertightness - see MIL-DTL-24643 for limits of water leakage ..... No failure

Group B:
- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C) 50
- Gas flame - 1 hour ..... No failure
- Drip - 95 ± 1°C ..... Zero
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35
- Cold bending, cable - (see Table I for mandrel diameter) ..... No damage

Physicals (unaged)
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

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Group C:
Physicals (aged)
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125
- Cable sealant removability ..... No failure
- Armor (type LS5KVTSGA only) - conformance to material, construction and coverage ..... No failure

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see Table I load current)	No failure
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5

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Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS: All types and sizes 1000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-020)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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